

DS701 Homework 2: Distances

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SOLUTIONS

Problem 1: The metric decides who is close

Practices: Minkowski distances, cosine similarity, Hamming and Jaccard — and noticing that “closest” depends on the choice.

Three short documents are represented as word-count vectors over the vocabulary (*data*, *model*, *error*):

$$\mathbf{A} = (1, 1, 1), \quad \mathbf{B} = (3, 3, 3), \quad \mathbf{C} = (6, 1, 1).$$

- (a) For each of the three pairs (A,B), (A,C), (B,C) compute the Minkowski distance $\|\mathbf{x} - \mathbf{y}\|_p$ for $p = 1$ (Manhattan), $p = 2$ (Euclidean) and $p = \infty$ (Chebyshev) — nine numbers, arranged as a 3×3 table. Under each of the three distances, which pair is **closest**? Note where the answer changes.
- (b) Compute the cosine similarity for the same three pairs (three decimals; $\sqrt{3} \approx 1.732$, $\sqrt{27} \approx 5.196$, $\sqrt{38} \approx 6.164$, $\sqrt{114} \approx 10.677$). Which pair is most similar under cosine? Explain in one sentence why cosine gives A and B the value it does even though every ℓ_p distance in (a) says they are some way apart, and say which of the two views is the right one if we want to know whether two documents are *about the same thing*.
- (c) Two pairs of shopping baskets, as sets of items:
 - $P = \{\text{bread, milk, eggs, butter, cheese, apples, rice, coffee}\}$ and $Q = \{\text{bread, milk, eggs, butter, cheese, apples, rice, tea}\}$;
 - $R = \{\text{bread, milk}\}$ and $S = \{\text{bread, jam}\}$.

For each pair compute the Hamming distance (the number of items in one basket but not the other, i.e. the size of the symmetric difference) and the Jaccard similarity $J_{\text{Sim}} = |x \cap y| / |x \cup y|$. Which pair does Hamming say is more alike, and which does Jaccard say? In one or two sentences, say which measure you would trust for “are these two shoppers similar?” and why.

Solution

- (a) Difference vectors: $\mathbf{A} - \mathbf{B} = (-2, -2, -2)$, $\mathbf{A} - \mathbf{C} = (-5, 0, 0)$, $\mathbf{B} - \mathbf{C} = (-3, 2, 2)$.

pair	$p = 1$	$p = 2$	$p = \infty$
(A,B)	6	$\sqrt{12} \approx 3.46$	2
(A,C)	5	$\sqrt{25} = 5$	5
(B,C)	7	$\sqrt{17} \approx 4.12$	3

Closest pair: **(A,C)** under Manhattan ($5 < 6 < 7$), but **(A,B)** under Euclidean (3.46) and under Chebyshev (2). The ranking changes with p : Manhattan adds up three moderate differences for (A,B) and penalizes them more than the single large difference of (A,C); Chebyshev sees only that single largest difference.

(b) $\cos(\mathbf{A}, \mathbf{B}) = \frac{3+3+3}{\sqrt{3}\sqrt{27}} = \frac{9}{\sqrt{81}} = 1.000$; $\cos(\mathbf{A}, \mathbf{C}) = \frac{6+1+1}{\sqrt{3}\sqrt{38}} = \frac{8}{\sqrt{114}} \approx \frac{8}{10.677} \approx 0.749$; $\cos(\mathbf{B}, \mathbf{C}) = \frac{18+3+3}{\sqrt{27}\sqrt{38}} = \frac{24}{\sqrt{1026}} \approx \frac{24}{32.03} \approx 0.749$ (same as (A,C), because $\mathbf{B} = 3\mathbf{A}$ has the same direction as $\mathbf{A}$). Most similar: **(A,B)**, with cosine exactly 1: $\mathbf{B} = 3\mathbf{A}$ points in the same direction, and cosine measures direction only — it is unchanged by rescaling either vector — while every ℓ_p distance measures the length of the difference vector, which is nonzero. For “about the same thing” cosine is the right view: B is A with every word used three times as often (a document three times as long on the same topic), and topic is a matter of word *proportions*, not counts.

(c) $P \cap Q$ has 7 items, $P \cup Q$ has 9, symmetric difference {coffee, tea}: Hamming = 2, $J_{\text{Sim}} = 7/9 \approx 0.778$. $R \cap S = \{\text{bread}\}$, $R \cup S = \{\text{bread, milk, jam}\}$, symmetric difference {milk, jam}: Hamming = 2, $J_{\text{Sim}} = 1/3 \approx 0.333$. Hamming calls the two pairs *equally* alike (2 and 2); Jaccard says (P,Q) is far more alike (0.78 vs 0.33). Trust Jaccard: P and Q agree on 7 of 9 items and differ on one swap, whereas R and S agree on 1 of 3 — Jaccard normalizes the disagreement by how much there was to agree on, Hamming counts raw disagreements regardless of basket size.

Common mistake: in (a), forgetting the square root for $p = 2$ (reporting 12, 25, 17), or taking max of the raw coordinates rather than of the absolute differences for $p = \infty$.

Problem 2: Is it a metric?

Practices: the metric axioms, especially the triangle inequality, and why they matter downstream.

Recall the four metric axioms: $d(x, x) = 0$; $d(x, y) > 0$ for $x \neq y$; $d(x, y) = d(y, x)$; and the triangle inequality $d(x, y) \leq d(x, z) + d(z, y)$ for all x, y, z .

- Let $P = (0, 0)$, $Q = (3, 4)$, $R = (6, 0)$ in $\mathbb{R}^2$. Compute the three Euclidean distances $d(P, Q)$, $d(Q, R)$, $d(P, R)$ and verify the triangle inequality for this triple in all three ways (each distance is at most the sum of the other two). Then state, in one line, when the triangle inequality holds with **equality** for Euclidean distance, and give a triple of points that achieves it.
- Cosine *distance* is $d_{\cos}(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y})$. The lecture says it is **not** a metric. Prove it: give three explicit vectors $\mathbf{x}, \mathbf{z}, \mathbf{y}$ in $\mathbb{R}^2$ for which $d_{\cos}(\mathbf{x}, \mathbf{y}) > d_{\cos}(\mathbf{x}, \mathbf{z}) + d_{\cos}(\mathbf{z}, \mathbf{y})$, and show the numbers (use $\cos 45^\circ = 1/\sqrt{2} \approx 0.707$ if it helps). *Hint:* think about angles 0° , 45° , 90° .
- Cosine distance fails a **second** axiom as well. Which one, and what is a two-vector example?

(One line.)

- (d) In two or three sentences: why does it matter, for nearest-neighbor search or for clustering, whether the dissimilarity you use satisfies the triangle inequality? Concretely, what can you conclude about $d(q, b)$ from knowing $d(q, a)$ is small and $d(a, b)$ is large when d is a metric — and what goes wrong when it is not?

Solution

- (a) $d(P, Q) = \sqrt{9+16} = 5$; $d(Q, R) = \sqrt{9+16} = 5$; $d(P, R) = \sqrt{36+0} = 6$. Checks: $d(P, R) = 6 \leq 5 + 5 = 10$; $d(P, Q) = 5 \leq 6 + 5 = 11$; $d(Q, R) = 5 \leq 5 + 6 = 11$. All hold (strictly). Equality holds exactly when the middle point lies **on the straight segment** between the other two (collinear, z between x and y), e.g. $P = (0, 0)$, $Z = (3, 4)$, $Y = (6, 8)$: $d(P, Y) = 10 = 5 + 5$.
- (b) Take $\mathbf{x} = (1, 0)$, $\mathbf{z} = (1, 1)$, $\mathbf{y} = (0, 1)$: angles 0° , 45° , 90° . $d_{\cos}(\mathbf{x}, \mathbf{y}) = 1 - 0 = 1$; $d_{\cos}(\mathbf{x}, \mathbf{z}) = 1 - \frac{1}{1 \cdot \sqrt{2}} = 1 - 0.707 = 0.293$, and by symmetry $d_{\cos}(\mathbf{z}, \mathbf{y}) = 0.293$. So $d_{\cos}(\mathbf{x}, \mathbf{z}) + d_{\cos}(\mathbf{z}, \mathbf{y}) = 0.586 < 1 = d_{\cos}(\mathbf{x}, \mathbf{y})$: going through $\mathbf{z}$ is a “shortcut”, which a metric forbids. One violating triple is a complete proof. (Remark: the *angle* itself, $\theta = \arccos(\cos)$, does satisfy the triangle inequality — here $90^\circ \leq 45^\circ + 45^\circ$; it is $1 - \cos$ that bends the scale.)
- (c) Positivity: $d(x, y) > 0$ for $x \neq y$ fails, since any two vectors in the same direction have $d_{\cos} = 0$ — e.g. $(1, 0)$ and $(3, 0)$: $\cos = 3/(1 \cdot 3) = 1$, so $d_{\cos} = 0$ although the vectors differ.
- (d) With a metric, small $d(q, a)$ and large $d(a, b)$ force $d(q, b) \geq d(a, b) - d(q, a)$ to be large: “close to something far from b ” means “far from b ”, so a search can skip b without computing $d(q, b)$, and a cluster of mutually close points cannot contain a point far from all its members. Without the triangle inequality none of that follows — q can be close to a , a close to b , and q far from b — so pruning is unsafe and “clusters” of near neighbors need not be internally coherent. (Cosine distance is still used constantly; the point is to know which guarantees you have given up.)

Common mistake: trying to prove (b) with vectors like $(1, 0)$ and $(2, 0)$ — those have $d_{\cos} = 0$ and show positivity failing (part (c)), not the triangle inequality.

Problem 3: Same points, different units, different neighbor

Practices: why raw Euclidean distance is dominated by the largest-range feature, and what standardizing does.

A bank describes customers by two features: annual income (in dollars) and age (in years). A query customer q and four customers on file:

	income	age
q	50,000	30
P_1	52,000	60
P_2	60,000	32
P_3	45,000	28
P_4	51,000	45

income	age
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To **standardize** a feature, replace each value v by its z -score $(v - \mu)/\sigma$, where μ and σ are the feature's mean and standard deviation. Use these values, taken from the bank's full customer base: income $\mu = 50,000$, $\sigma = 10,000$; age $\mu = 40$, $\sigma = 10$.

- Using raw Euclidean distance, which of $P_1, \dots, P_4$ is the nearest neighbor of q ? (Squared distances are enough to rank; report them.) In one line: how much does the age difference contribute to any of these distances?
- Standardize all five rows with the given μ, σ (write the z -scores), recompute the four squared Euclidean distances from q , and give the nearest neighbor now. Does it change?
- Redo (a) with income measured in *thousands* of dollars (50, 52, 60, 45, 51) and age unchanged. Which neighbor wins? Then answer: which of the three answers should a data scientist trust as “the customer most like q ”, and why — what is wrong with the other two?
- Standardizing is not always right. Give one situation where you would **not** standardize before computing distances, and one sentence on why (think about features that are already in the same meaningful units, or a feature whose spread *should* count).

Solution

- Squared raw distances from q : P_1 : $2000^2 + 30^2 = 4,000,000 + 900 = 4,000,900$; P_2 : $10,000^2 + 2^2 = 100,000,004$; P_3 : $5000^2 + 2^2 = 25,000,004$; P_4 : $1000^2 + 15^2 = 1,000,225$. Nearest neighbor: P_4 ($d \approx 1000.1$). Age contributes at most 900 out of millions — well under 0.1%; the ranking is decided by income alone.
- z -scores $((\text{income} - 50,000)/10,000, (\text{age} - 40)/10)$: $q = (0, -1)$, $P_1 = (0.2, 2)$, $P_2 = (1, -0.8)$, $P_3 = (-0.5, -1.2)$, $P_4 = (0.1, 0.5)$. Squared distances from q : P_1 : $0.04 + 9 = 9.04$; P_2 : $1 + 0.04 = 1.04$; P_3 : $0.25 + 0.04 = 0.29$; P_4 : $0.01 + 2.25 = 2.26$. Nearest neighbor: P_3 ($d = \sqrt{0.29} \approx 0.54$). Yes — it flips: P_4 , the raw winner, is now third, because its 15-year age gap (1.5 SD) finally counts.
- Income in thousands: P_1 : $2^2 + 30^2 = 904$; P_2 : $10^2 + 2^2 = 104$; P_3 : $5^2 + 2^2 = 29$; P_4 : $1^2 + 15^2 = 226$. Nearest: P_3 — a *third* computation on the same customers, and now age dominates instead of income. Trust (b). The raw answers in (a) and (c) depend on an arbitrary bookkeeping choice — dollars vs thousands of dollars — that says nothing about customers; each lets whichever feature happens to have the bigger numbers decide. Standardizing puts both features on the same scale (SD units), so a 1-SD income gap and a 1-SD age gap count equally and the answer no longer changes when someone re-enters the spreadsheet in different units.
- Any of: (i) features already in the *same* meaningful units where spread is information — e.g. the pixel intensities of an image, or coordinates in meters, or the same measurement at 12 months: standardizing would inflate a nearly-constant, mostly-noise feature to equal importance with a genuinely varying one; (ii) one-hot/binary indicator features, whose 0/1 scale is already comparable; (iii) when the domain says one feature *should* dominate. The rule: standardize when the units are arbitrary and incomparable; don't when the units are shared and the spread is the signal.

Common mistake: standardizing with the mean and SD of only the five rows in the table when the problem gives population values — either is defensible only if you say which you used; here the given values are the ones to use.